

Thrinayani Yedhoti

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SUMMARY

MS Computer Science candidate at University of Washington (GPA: 3.83) with 1.5+ years of backend and full-stack engineering experience shipping production software at Schneider Electric R&D (industrial automation). Built REST APIs, backend services, dashboards, and access-control systems for manufacturing plant operations. Seeking Summer/Fall 2026 Software Engineer Internship in industrial automation or autonomous manufacturing.

SKILLS

- Languages:** Python, TypeScript, Java, C#, C++, JavaScript, SQL
- Backend & APIs:** FastAPI, Spring Boot, ASP.NET, Node.js, REST APIs, .NET, PostgreSQL, MySQL, Redis, SQLite, SQLAlchemy (ORM)
- Dashboards & Frontend:** Angular, WPF, ASP.NET, RBAC, OAuth
- Data & Automation:** Real-time edge computing, CI/CD, Docker, Docker Compose, Azure DevOps
- AI/ML:** PyTorch, CNNs, Spiking Neural Networks, Computer Vision, OpenCV, LLM Integration (Copilot, Claude)
- Methods:** Agile/SCRUM, Unit Testing, Code Review, System Design, Large Codebase Navigation, SonarQube, Coverity

EDUCATION

MS, Computer Science and Software Engineering **Expected: March 2027**
University of Washington, Bothell *GPA: 3.83/4.0*

- **Coursework:** Advanced Computer Vision, Algorithm Analysis & Design, Research Methods in SE, Evaluating Software Design | **CS Grader:** Introduction to AI (Jan–Mar 2026)

EXPERIENCE

Schneider Electric – R&D, Industrial Automation **Feb 2025 – Aug 2025**
Software Design Engineer *Bengaluru, India*

- Reduced PLC license validation time by 30% by engineering a GSE mechanism for Unity M580 industrial controllers using **C#** and **WPF** with full unit test coverage, directly supporting factory automation workflows.
- Decreased carbon emission reporting latency for manufacturing plants by architecting a real-time **edge computing pipeline** using **Python** and **FastAPI**, improving operational visibility into production metrics.
- Accelerated vulnerability inspection workflows by building **user management dashboards** and RBAC-based access controls using **Angular** and **ASP.NET**.

Schneider Electric – R&D, Industrial Automation **Aug 2024 – Feb 2025**
Graduate Engineer Trainee *Bengaluru, India*

- Reduced application load times by 20% by refactoring **.NET**, **WPF**, and **Angular** codebases supporting industrial automation software, with maintained unit test coverage across all modules.
- Accelerated database query execution by 40% by designing an **ORM-based data architecture** using **SQLite** and **SQLAlchemy**.

Schneider Electric – R&D, Industrial Automation **Jan 2024 – Jul 2024**
Software Engineer Intern *Bengaluru, India*

- Shipped production-ready **RBAC system** using **C#**, **Angular**, and **.NET** with zero vulnerabilities across Sonar and Coverity static analysis; delivered clean, well-tested code reviewed by senior engineers.

PROJECTS

Distributed Rate Limiter Microservice (Java, Spring Boot, Redis, PostgreSQL, Docker) | [Link](#) **May 2026**

- Built production-ready distributed rate-limiting **backend service** in **Java** and **Spring Boot 3** implementing token bucket and sliding window algorithms via **Redis**; deployed live via **Docker Swarm** with **CI/CD** through GitHub Actions, demonstrating end-to-end ownership of a real production system.
- Persisted full audit trail to **PostgreSQL** using **Hibernate/JPA**; exposed 3 **REST API** endpoints returning HTTP 429 on threshold breach with per-client stats, providing operational visibility into API usage and capacity.

Optimal Seamlines for Image Stitching via Fast Marching Method (C++, OpenCV) | [Link](#) **Nov–Dec 2025**

- Outperformed GraphCutSeamFinder and DPSeamFinder in suppressing parallax artifacts by implementing an FMM-based seam finder in an **OpenCV** pipeline; collaborated cross-functionally with Lawrence Sanchez (Engineering Director, Minecraft/Microsoft).
- Solved Eikonal equation over fused cost fields (color, edge, texture, saliency) to compute globally optimal seam paths, delivering stable runtime and measurable quality improvements across benchmark datasets.

PUBLICATIONS

- [Attention-Based Spiking Neural Networks for Fashion Product Labelling](#) — *IEEE, 2024*
- [CNN-Based Age Estimation using Diverse Facial Datasets](#) — *IEEE, 2024*